

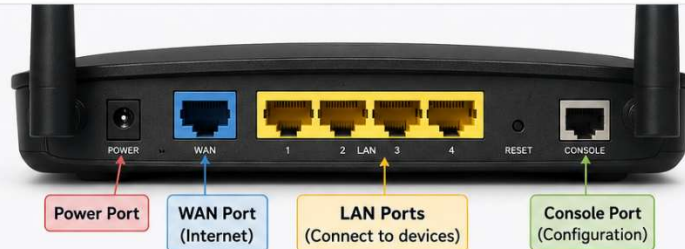
1.Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Device	Main Function	Typical Ports
Switch	Connects many devices in the same local network.	Ethernet ports, Console port, Power
Router	Connects the local network to the Internet and sends data between networks.	WAN port, LAN Ethernet ports, Console, Power
Hub	Connects multiple devices and sends data to all connected devices.	Ethernet ports, Power
Firewall	Protects the network by blocking unwanted or dangerous traffic.	WAN, LAN Ethernet ports, Console, Power
Access Point (AP)	Provides Wi-Fi connection to wireless devices.	Ethernet port, Power, Antenna

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

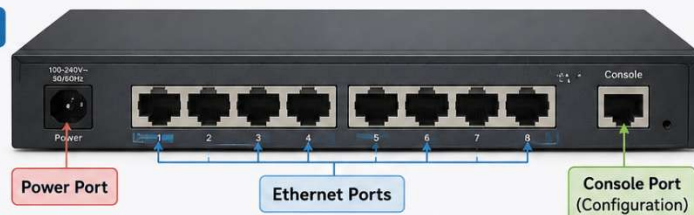
NETWORK DEVICES AND THEIR PORTS

1. ROUTER



- Connects your home/office network to the Internet.
- Assigns IP addresses.
- Routes data between networks.

2. SWITCH



- Connects multiple devices in the same network.
- Forwards data to the correct device using MAC address.
- Used in homes, offices, schools, etc.

3. ACCESS POINT (AP)



- Provides Wi-Fi (wireless) connection.
- Connects wireless devices (phones, laptops, tablets) to the network.
- Usually connected to router or switch using Ethernet.

3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Example: Using Instagram

When I use Instagram on my phone, the Access Point connects my phone to the Wi-Fi network. The Router sends my data from my home network to the Internet and helps it reach Instagram's servers. Switches can be used inside the network or data center to connect servers and network devices. Firewalls help protect the network by checking and blocking unwanted traffic.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

- One possible reason is that the Access Point (AP) is handling too many students and devices at the same time. This can cause slow speed and frequent disconnections.
- Solution: The college can upgrade to a newer and higher-capacity Wi-Fi 6/6E access point or add more access points. This can provide better coverage and handle more users at the same time, improving Wi-Fi speed and stability.